ChestNet: A Compact Convolutional Network for Multilabel Thoracic Finding Classification

Taher Akolawala

Abstract—We developed ChestNet, a compact convolutional neural network for predicting 14 thoracic findings from 28×28 grayscale chest radiographs in ChestMNIST. The model combines three convolutional blocks with a sigmoid multilabel head and uses validation-area-under-the-curve (AUC) checkpointing and early stopping. On the 22,433-image test split, ChestNet achieved a macro AUC of 0.7353. At a 0.5 threshold it rarely predicts any finding: recall is zero for 11 of 14 findings and highest for effusion at 7.8%. Its 94.77% binary accuracy matches the 94.74% obtained by predicting every label negative, so accuracy is uninformative here. ChestNet provides an exploratory baseline for low-resolution multilabel classification; the results do not establish clinical utility.

Index Terms—chest radiography, ChestMNIST, convolutional neural network, multilabel classification, class imbalance

I. Introduction

Chest radiographs can contain several findings at once, so thoracic finding detection is naturally a multilabel problem. A model must assign an independent score to each finding rather than force all images into one mutually exclusive category. We developed ChestNet to study this setting with a compact network on a low-resolution benchmark. The objective was to train and evaluate a 14-output classifier and examine the gap between aggregate scores and per-finding behavior. The source dataset comes from NIH ChestX-ray14 through the standardized MedMNIST v2 ChestMNIST release [1], [2].

This distinction matters because a single summary metric can obscure false negatives. When negative labels dominate, predicting absence frequently yields high elementwise accuracy. We therefore examine discrimination and thresholded confusion matrices alongside binary accuracy, while treating missed positive examples as a central limitation.

II. Dataset and Preprocessing

ChestMNIST contains 112,120 frontal chest radiographs with 14 binary finding labels. Its fixed split has 78,468 training, 11,219 validation, and 22,433 test images [1]. Each image is supplied at 28×28 pixels. We scaled grayscale intensities from $[0, 255]$ to $[0, 1]$ and expanded each image tensor from $(28, 28)$ to $(28, 28, 1)$ for two-dimensional convolution. The experiment used the provided benchmark split; it did not include an external cohort, high-resolution images, or prospective clinical cases.

For an image x and labels $y \in \{0, 1\}^{14}$, the network produces probabilities $p_k(x)$ for each finding k. Training minimizes binary cross-entropy across the output labels:

$$\mathcal{L} = -\frac{1}{14} \sum_{k=1}^{14} \left[y_k \log p_k + (1 - y_k) \log(1 - p_k) \right]. \qquad (1)$$

This objective permits co-occurring findings, unlike a softmax classifier. It does not by itself compensate for unequal finding prevalence.

III. Model and Training

ChestNet has three 3×3 convolutional blocks with 32, 64, and 128 filters. Each block includes batch normalization and max pooling. After flattening, a 256-unit dense layer and 0.5 dropout feed 14 sigmoid outputs. The resulting network has 392,334 parameters, of which 391,886 are trainable. The 28×28 input limits the spatial detail available for small abnormalities.

We optimized binary cross-entropy with Adam at an initial learning rate of 10^{-3} for at most 50 epochs. We retained the checkpoint with the best validation AUC, reduced the learning rate by a factor of 0.2 after a plateau, and used early stopping with a 10-epoch patience. Training stopped after 20 epochs; the selected checkpoint was from epoch 10. Subsequent divergence between the training and validation curves was consistent with overfitting.

IV. Results and Interpretation

TABLE I
ChestNet test-set results

Measure	ChestNet	All negative
Macro AUC	0.7353	0.5
Macro recall (0.5 threshold)	0.007	0
Binary accuracy	94.77%	94.74%
Binary cross-entropy loss	0.1667	–

Table I summarizes ChestNet's test results. Binary accuracy counts correct decisions across all image–label pairs. Only 16,510 of the 314,062 test pairs are positive, so a classifier that predicts every label negative scores 94.74%, within 0.03 points of ChestNet. The AUC of 0.7353 provides a threshold-independent view of discrimination, though an aggregate AUC still does not describe performance for every finding.

Per-finding confusion matrices at a 0.5 threshold show almost no positive predictions. Only four findings have any true positives: effusion (216 of 2,754, 7.8% recall),

cardiomegaly (9 of 582), pneumothorax (1 of 1,089), and infiltration (2 of 3,938); the remaining ten have zero recall. This behavior rules out interpreting the model as a triage aid at this threshold, since missed positives could delay review. We did not measure precision–recall curves, validation-tuned thresholds, calibration, confidence intervals, or performance against a clinical baseline. Those evaluations are needed to assess operating points for individual findings.

V. Limitations and Next Steps

Low spatial resolution and uneven label frequencies are the principal constraints. More clinically meaningful experiments would use higher-resolution images, patient-level separation checks, and an external test cohort. Class-weighted or focal losses, balanced sampling, and data augmentation could be evaluated against the current baseline, but their benefits should be measured rather than assumed. Thresholds should be selected on validation data for an explicit sensitivity target and then fixed before test evaluation. Reporting recall, precision, AUC, and calibration separately for all 14 findings would expose failures that aggregate accuracy hides.

ChestNet has not been validated for diagnosis, worklist prioritization, or patient care. Before any such use, it would need robust external validation, subgroup analysis, and review of false negatives under realistic prevalence and workflow conditions. Our experiment illustrates why multilabel medical-image results require class-specific interpretation.

VI. Conclusion

ChestNet achieved 0.7353 test AUC on ChestMNIST with a 14-output CNN. Its 94.77% binary accuracy is no better than predicting every label negative, and at a 0.5 threshold it detects almost no positive findings. Future work should prioritize class-specific sensitivity and external validation before making operational claims.

References

[1] J. Yang et al., "MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," Scientific Data, vol. 10, art. 41, 2023.

[2] X. Wang et al., "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in Proc. IEEE CVPR, 2017, pp. 2097–2106.